

Breast Cancer Classification Convolutional Neural Network

Hailey Zweedyk

Grand Valley State University

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Prof. Kel Levick

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Introduction

This project aims to support medical image classification by detecting benign and malignant tumors in breast ultrasound images. Breast cancer is one of the most commonly diagnosed cancers, accounting for approximately 30% of all new cancer diagnoses among women in the United States. An estimated 321,910 women are diagnosed annually, and approximately 13% of women will develop breast cancer in their lifetime (American Cancer Society, 2024). Although significantly less common, men can also develop breast cancer, accounting for fewer than 1% of cases, with approximately 2,670 diagnoses each year.

Radiologist interpretation of breast imaging is highly accurate; however, carefully trained convolutional neural network (CNN) models can serve as support tools by improving consistency, assisting early detection, and identifying subtle image features that may be difficult to detect through visual assessment alone.

Dataset

The dataset used in this project is the Breast Ultrasound Images Dataset published by Al-Dhabyani et al. (2020). It contains 1,578 ultrasound images categorized into three classes: benign, malignant, and normal. This dataset does have noticeable class imbalance, but using transfer learning, this dataset should produce adequate diagnostic results.

Goal

Given the limited data available, the objective of this model is to achieve at least 85% recall, precision, F1 score, and accuracy, with malignant recall serving as the primary evaluation metric. Prioritizing recall for malignant cases helps reduce false negatives, which is especially important in medical screening contexts.

Model Setup

Transfer learning with ResNet50 will be used to develop a CNN classifier. The pretrained ResNet50 model extracts high-level image features and outputs 2048 feature channels. These features are aggregated using global average pooling, producing a 2048-dimensional feature vector.

A fully connected layer with 128 neurons and ReLU activation is applied to learn task-specific feature representations. A dropout rate of 30% is used for regularization to reduce overfitting. The final softmax layer outputs probability predictions across three classes: benign, malignant, and normal.

Process:

Image Resolution: 200x200

Training runs:

- Trainable = False & Optimizer = Adam: 10 epochs
- Trainable = True & Optimizer = Adam(1e-5): 10 epochs
- Weight Adjustments & Trainable = True [-40:] & Optimizer = Adam(1e-5): 10 epochs

Model Problems & Solutions

In early models, it achieved high accuracy and high precision; however, recall remained < 80%, specifically with malignant tumors being < 60%. This was due to weights being assigned only by the amount of data rather than the importance of the data. Changing the weights to normal = 1.2, malignant = 2.3, and benign = 1.0 yielded much higher recall amongst all categories with malignant recall reaching 82% after the first model training with the new weights. Testing

malignant = 2.5 resulted in significant performance decline, meaning 2.3 is the most optimal malignant weight.

Final Model

The final model achieved 90% recall for malignant cases, exceeding the target. Benign recall reached 94%, likely due to its majority representation in the dataset. In contrast, normal recall was lower at 79%, which can be attributed to its minority status and the class weighting favoring malignant cases. The model reached 91% overall accuracy, demonstrating strong performance for medical classification. With 30 epochs (10 frozen, 20 unfrozen) and custom class weights, it surpassed all models trained for 50+ epochs in malignant recall, likely because the dataset size leads to early overfitting.

Future Improvements

Future work should focus on increasing the dataset size, which could help mitigate class imbalance and improve overall model performance. Additionally, further experimentation with more finely tuned hyperparameters and additional model runs is recommended to ensure optimal model selection given the available data.

References

American Cancer Society. (2024, January 12). *Key statistics for breast cancer*. American Cancer Society.

<https://www.cancer.org/cancer/types/breast-cancer/about/how-common-is-breast-cancer.html>